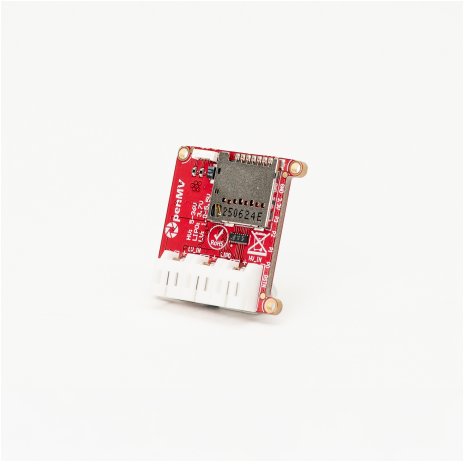
OpenMV® AE3 Battery Shield

Datasheet

SKU: OMV-AE3-BATTERY-SHIELD-V4

Description

The AE3 Battery Shield is a power and storage expansion board for the OpenMV AE3 camera. It provides flexible power input options including a 2.1 mm DC barrel jack accepting 1.8 V to 5.5 V (suitable for three alkaline batteries in series), a wide-range 6 V to 36 V input with reverse voltage protection for industrial and automotive applications, and a 3.7 V LiPo battery charger with 100 mA charging from either the wide-range input or 5 V USB. The board also includes a SPI-attached microSD card socket for data logging and storage, addressing the AE3's lack of on-board microSD. The shield mounts directly on the back of the AE3 via its board-to-board connector.



Target Areas

Battery-powered deployments, remote sensing, field installations, data logging, solar-powered systems, industrial monitoring, wildlife cameras, agricultural sensing.

Key Features

- **Low-voltage input** — 1.8–5.5 V via 2.1 mm DC barrel jack. Supports three alkaline batteries in series.
- **Wide-range input** — 6–36 V input with reverse voltage protection for industrial, automotive, and solar applications.
- **LiPo charger** — 3.7 V single-cell LiPo battery charger at 100 mA from wide-range input or 5 V USB.
- **microSD storage** — SPI-attached microSD card socket for data logging, image storage, and model files.
- **Ultra-lightweight** — Only 9 g, adding minimal weight to the AE3 system.

Applications

The AE3 Battery Shield is ideal for battery-powered deployments, remote sensing, field installations, data logging, solar-powered systems, and industrial monitoring where flexible power input and removable storage are needed.

Contents

1 Features and System Components
- 1.1 Power Input
- 1.2 Storage
- 1.3 Compatibility

2 Pinout
- 2.1 microSD Card Pins

3 General Info

4 Power
- 4.1 Power Input Specifications
- 4.2 Power Consumption

5 Company Information

6 Reference Documentation

7 Revision History

1 Features and System Components

1.1 Power Input

Component	Details
LV PWR	2.1 mm DC barrel jack. Accepts 1.8 V to 5.5 V input. Reverse voltage tolerant to -5.5 V. Compatible with three alkaline batteries in series (3x 1.5 V = 4.5 V nominal). Provides 3.0 V output, up to 800 mA draw.
HV PWR	6 V to 36 V wide-range input. Reverse voltage tolerant to -36 V. Suitable for 12 V automotive systems, 24 V industrial power, and solar panel direct connection. Provides 3.3 V output, up to 600 mA draw.
BAT	3.7 V 1S LiPo battery input. NOT reverse voltage tolerant. Charges at up to 100 mA from HV or USB inputs. Provides 3.0 V output, up to 800 mA draw.

1.2 Storage

Component	Details
microSD Card	SPI-attached microSD card socket on P0–P3 (MOSI, MISO, SCLK, SS) with power enable on P11. ~300 KB/s write, ~100 KB/s read. Provides removable storage for data logging, captured images, trained models, and MicroPython scripts.

1.3 Compatibility

Component	Details
OpenMV AE3	Designed for the OpenMV AE3 camera module. Mounts directly on the back of the AE3 via board-to-board connector.

2 Pinout

2.1 microSD Card Pins

Pin	Function
P0	MOSI
P1	MISO
P2	SCLK
P3	SS
P11	Power Enable

3 General Info

Dimension	Value
Length	32 mm
Width	32 mm
Height	12 mm
Weight	9 g
Operating Temperature	-25°C to +85°C
Storage Temperature	-25°C to +85°C
Country of Origin	China
HS Code	8473.30.1180
UPC	602772227229
SKU	OMV-AE3-BATTERY-SHIELD-V4
ECCN	EAR99

4 Power

4.1 Power Input Specifications

Input	Voltage Range	Output Rail	Max Draw	Notes
LV PWR	1.8–5.5 V	3.0 V	800 mA	Reverse voltage tolerant to -5.5 V
HV PWR	5–36 V	3.3 V	600 mA	Reverse voltage tolerant to -36 V
BAT	~3.7 V	3.0 V	800 mA	NOT reverse voltage tolerant

All external interfaces have robust transient voltage protection.

4.2 Power Consumption

Condition	Current Draw	Input
Idle (not charging)	2 mA	HV or USB @ 5 V
Charging LiPo	130 mA	HV or USB @ 5 V
No load (deep sleep)	16 µA	LV @ 5 V
No load (deep sleep)	12 µA	BAT @ 3.7 V

5 Company Information

Field	Details
Website	https://openmv.io
Documentation	https://docs.openmv.io
Support	https://forums.openmv.io

6 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/openmv-ae3-battery-shield
3D Mounts	https://grabcad.com/library/openmv-ae3-with-a-shield-mounts-1

7 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.